Pidamarthi Praveen Kumar

praveenkumarpidamarthi01@gmail.com — 8317668826 — GitHub

EDUCATION

Chaitanya Bharathi Institute of Technology Nov 2022 – Aug 2026
B.Tech in Artificial Intelligence and Machine Learning CGPA: 7.11

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS
AI & Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Natural Language Processing (NLP), Embeddings, Vector Search
Machine Learning & Deep Learning: Machine Learning, Deep Learning, CNN, BiLSTM, Sequence Modeling, Model Evaluation
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, SentenceTransformers, NumPy, Pandas, OpenCV, Flask, Tkinter
Web Development: HTML, CSS, JavaScript, Flask, React JS, Responsive Web Design
Databases & Backend: SQL, SQLite, REST APIs, Client-Server Architecture, Socket Programming
Core CS: Data Structures, Algorithms, Object-Oriented Programming, Computer Networks
Tools: Git, GitHub, VS Code, Google Colab

PROJECTS

Personal Portfolio Website GitHub
HTML, CSS, JavaScript

- Designed and developed a responsive personal portfolio website to showcase projects, technical skills, and profile.
- Built the website using semantic HTML, custom CSS, and JavaScript for an interactive user experience.
- Created dedicated sections for education, skills, projects, certifications, and contact information.
- Maintained the complete portfolio source code and assets in a public GitHub repository.

Praveen Coding Asst & RAG Chatbot GitHub
Python, Flask, Qwen 2.5 Coder 7B, SentenceTransformers, RAG

- Built an AI-powered coding assistant capable of explaining Python, SQL, DSA, algorithms, and AI/ML concepts.
- Integrated Qwen 2.5 Coder 7B as the underlying large language model for coding-focused responses.
- Implemented PDF-based Retrieval-Augmented Generation using SentenceTransformers embeddings and cosine similarity.
- Added document retrieval with cited source pages and real-time token streaming for interactive responses.
- Developed a responsive web interface using Flask, HTML, CSS, JavaScript, and Markdown rendering.

EEG-to-Text Conversion System GitHub
Python, PyTorch/TensorFlow, Deep Learning

- Built an end-to-end deep learning pipeline to convert raw EEG signals into textual output.
- Designed a hybrid CNN-BiLSTM architecture to extract spatial and temporal features from EEG signals.
- Preprocessed EEG data through filtering, normalization, and segmentation to improve model performance.
- Applied sequence modeling techniques to map neural signal patterns to textual predictions.
- Evaluated model performance using BLEU score and loss-curve analysis.

PulseChat GitHub
Python, Tkinter, Socket Programming, SQLite

- Developed a real-time multi-user chat application using Python socket programming and a client-server architecture.
- Implemented concurrent client handling using Python threading for simultaneous user connections.
- Built login and signup interfaces with user authentication and SQLite-based credential management.
- Developed a graphical chat interface with online-user tracking and real-time message broadcasting.
- Integrated separate server, client, database, and GUI components for modular application development.

RELEVANT COURSEWORK

Web Development (Internshala) • AI Foundation and Advanced (ISB) • Machine Learning Foundation (ISB) • Machine Learning (Udemy)